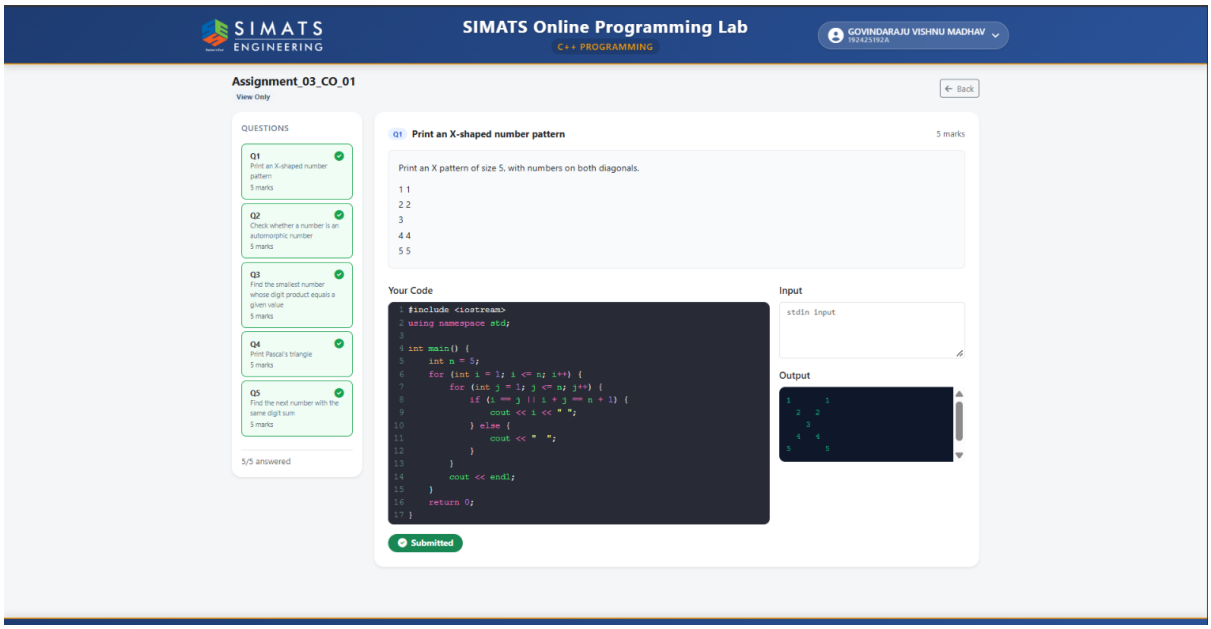
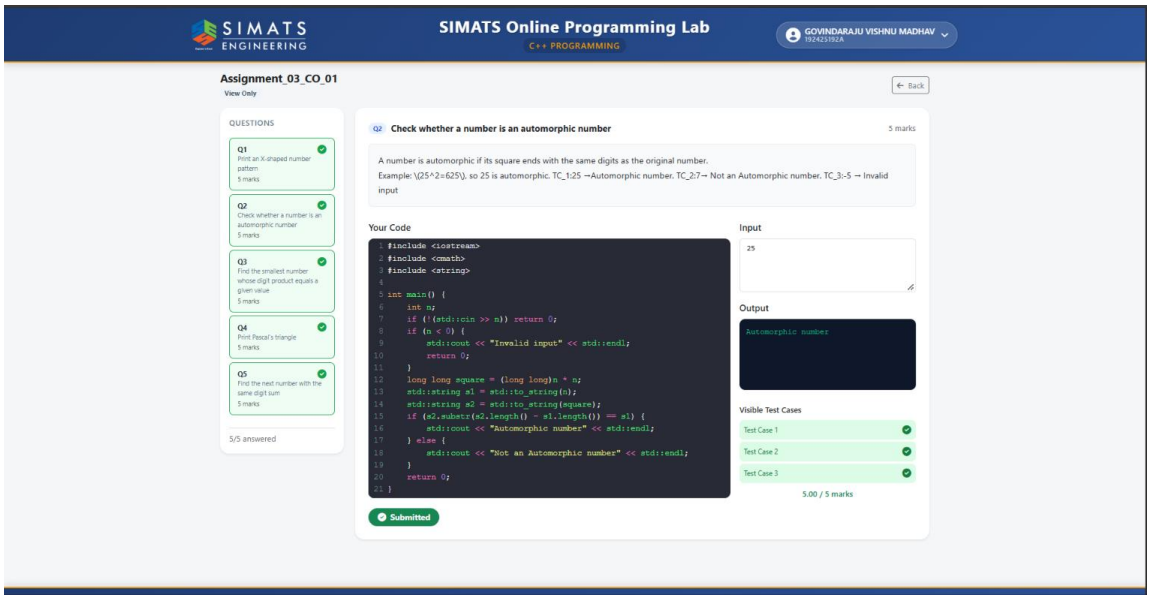
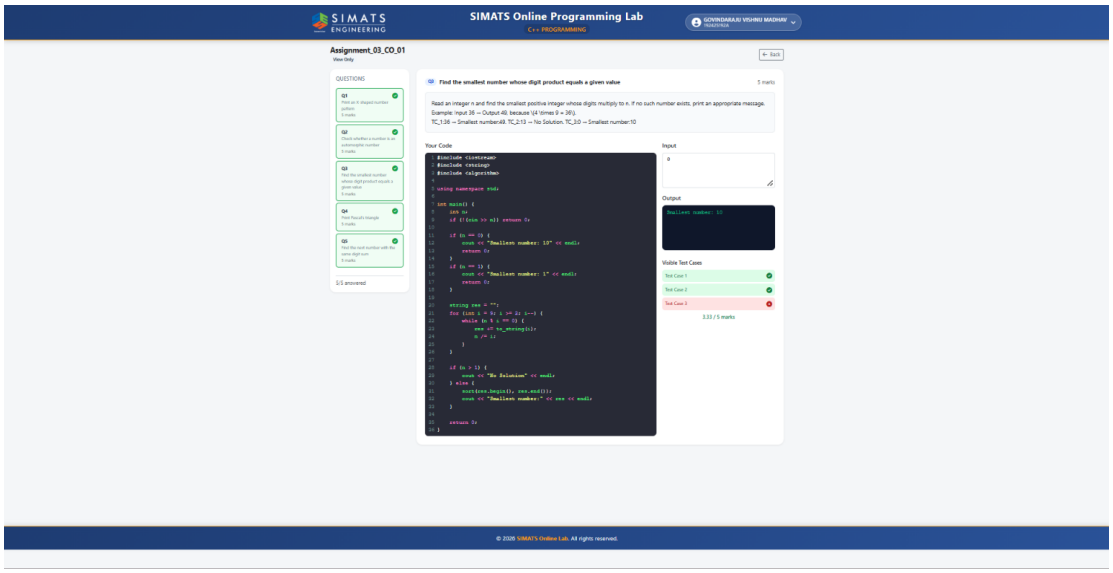
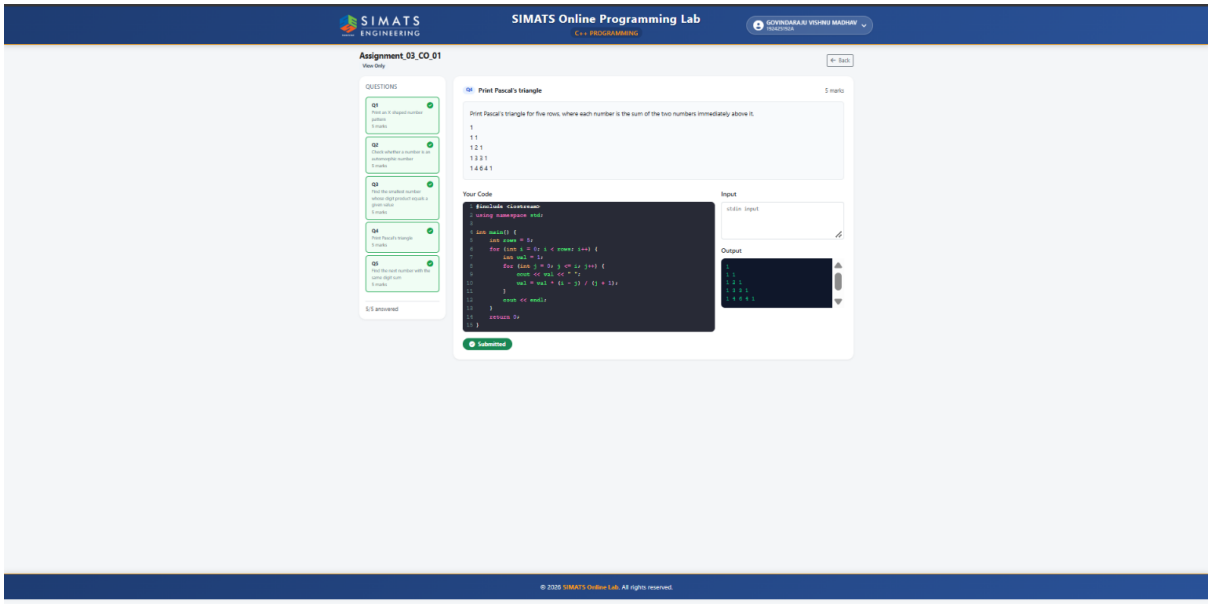
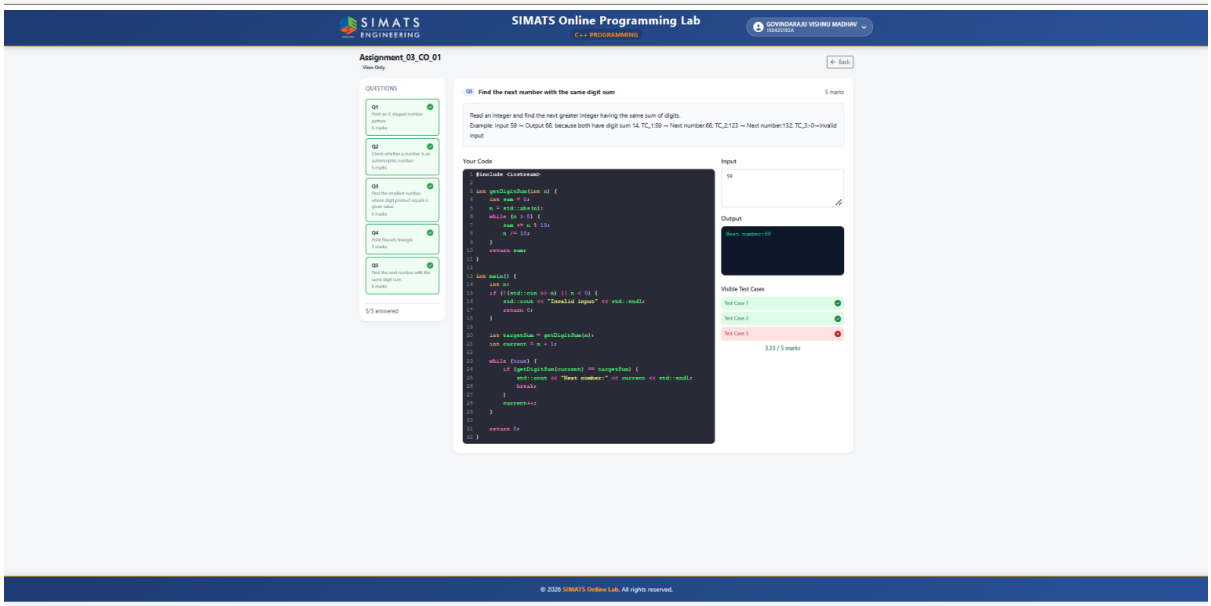
Co1 At3


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QUESTIONS
Q1 Print an X-shaped number pattern
Q2 Check whether a number is an automorphic number
Q3 Find the smallest number whose digit product equals a given value
Q4 Print Pascal's triangle
Q5 Find the next number with the same digit sum
5/5 answered
Q1 Print an X-shaped number pattern
5 marks
Print an X pattern of size 5, with numbers on both diagonals.
1 1
2 2
3
4 4
5 5
Your Code
#include <iostream>
using namespace std;
int main() {
int n = 5;
for (int i = 1; i <= n; i++) {
for (int j = 1; j <= n; j++) {
if (i == j || i + j == n + 1) {
cout << i << " ";
} else {
cout << " ";
}
}
cout << endl;
}
return 0;
}
Input
stdin Input
Output
Submitted


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QUESTIONS
Q1 Print an X-shaped number pattern
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Q3 Find the smallest number whose digit product equals a given value
Q4 Print Pascal's triangle
Q5 Find the next number with the same digit sum
5/5 answered
Q2 Check whether a number is an automorphic number
5 marks
A number is automorphic if its square ends with the same digits as the original number.
Example: \(25^2=625\), so 25 is automorphic. TC_1:25 →Automorphic number. TC_2:7→ Not an Automorphic number. TC_3:-5 → Invalid input
Your Code
#include <iostream>
#include <cmath>
#include <string>
int main() {
int n;
if (!(std::cin >> n)) return 0;
if (n < 0) {
std::cout << "Invalid input" << std::endl;
return 0;
}
long long square = (long long)n * n;
std::string s1 = std::to_string(n);
std::string s2 = std::to_string(square);
if (s2.substr(s2.length() - s1.length()) == s1) {
std::cout << "Automorphic number" << std::endl;
} else {
std::cout << "Not an Automorphic number" << std::endl;
}
return 0;
}
Input
25
Output
Automorphic number
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
5.00 / 5 marks
Submitted


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QUESTIONS
Find the smallest number whose digit product equals a given value
5 marks
Your Code
Input
Output
Visible Test Cases
3.33 / 5 marks
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QUESTIONS
Q1
Q2
Q3
Q4
Q5
5/5 answered
Q4 Print Pascal's triangle
5 marks
Print Pascal's triangle for five rows, where each number is the sum of the two numbers immediately above it.
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
Your Code
#include <iostream>
using namespace std;
int main() {
int rows = 5;
for (int i = 0; i < rows; i++) {
int val = 1;
for (int j = 0; j <= i; j++) {
cout << val << " ";
val = val * (i - j) / (j + 1);
}
cout << endl;
}
return 0;
}
Input
stdin input
Output
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
Submitted
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QUESTIONS
Q1
Q2
Q3
Q4
Q5
5/5 answered
Q5 Find the next number with the same digit sum
5 marks
Read an integer and find the next greater integer having the same sum of digits.
Example: Input 59 → Output 68, because both have digit sum 14. TC_1:59 → Next number:68. TC_2:123 → Next number:132. TC_3:-0→Invalid input
Your Code
#include <iostream>
int getDigitSum(int n) {
int sum = 0;
n = std::abs(n);
while (n > 0) {
sum += n % 10;
n /= 10;
}
return sum;
}
int main() {
int n;
if (!(std::cin >> n) || n < 0) {
std::cout << "Invalid input" << std::endl;
return 0;
}
int targetSum = getDigitSum(n);
int current = n + 1;
while (true) {
if (getDigitSum(current) == targetSum) {
std::cout << "Next number:" << current << std::endl;
break;
}
current++;
}
return 0;
}
Input
59
Output
Next number:68
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
3.33 / 5 marks
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